八 二重积分

一、选择题

1. (C). $\lim\limits_{n\to\infty}\dfrac{1}{n^2}\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{n}f\left(\dfrac{i}{n},\dfrac{2j-1}{n}\right)=\lim\limits_{n\to\infty}\dfrac{1}{n^2}\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{n}f\left(\dfrac{i}{n},\dfrac{2j}{n}\right)=\int_0^1\mathrm{d}x\int_0^1 f(x,2y)\mathrm{d}y$

$$=\frac{1}{2}\int_0^1\mathrm{d}x\int_0^1 f(x,2y)\mathrm{d}(2y)=\frac{1}{2}\int_0^1\mathrm{d}x\int_0^2 f(x,y)\mathrm{d}y.$$

选(C).

2. (B). 由积分中值定理,存在$(\xi,\eta)\in D$,使得

$$\iint_D \mathrm{e}^{-x^2}\cos 2y\,\mathrm{d}\sigma=\mathrm{e}^{-\xi^2}\cos 2\eta\cdot\pi\cdot t\cdot\frac{t}{2};$$

再由$1-\mathrm{e}^{-\frac{t^2}{4}}\sim\dfrac{t^2}{4}(t\to 0^+)$,得$\lim\limits_{t\to 0^+}\dfrac{\iint\limits_D \mathrm{e}^{-x^2}\cos 2y\,\mathrm{d}\sigma}{1-\mathrm{e}^{-\frac{t^2}{4}}}=2\pi\lim\limits_{t\to 0^+}\mathrm{e}^{-\xi^2}\cos 2\eta=2\pi$. 选(B).

3. (C). 令$\begin{cases}x=r\cos\theta,\\ y=r\sin\theta,\end{cases}$则

$$\iint_D f(a-\sqrt{x^2+y^2})\mathrm{d}x\,\mathrm{d}y=2\pi\int_0^a rf(a-r)\mathrm{d}r\xlongequal{a-r=u}2\pi\left[a\int_0^a f(u)\mathrm{d}u-\int_0^a uf(u)\mathrm{d}u\right],$$

$\iint\limits_D\cos(x+y^2)\mathrm{d}x\,\mathrm{d}y=\pi a^2\cdot\cos(\xi+\eta^2)$,其中$(\xi,\eta)\in D$,则

$$\lim_{a\to 0^+}\frac{\iint\limits_D f(a-\sqrt{x^2+y^2})\mathrm{d}x\,\mathrm{d}y}{a\iint\limits_D\cos(x+y^2)\mathrm{d}x\,\mathrm{d}y}=2\lim_{a\to 0^+}\frac{1}{\cos(\xi+\eta^2)}\cdot\frac{a\int_0^a f(u)\mathrm{d}u-\int_0^a uf(u)\mathrm{d}u}{a^3}$$

$$=2\lim_{a\to 0^+}\frac{a\int_0^a f(u)\mathrm{d}u-\int_0^a uf(u)\mathrm{d}u}{a^3}$$

$$=\frac{2}{3}\lim_{a\to 0^+}\frac{\int_0^a f(u)\mathrm{d}u}{a^2}=\frac{1}{3}\lim_{a\to 0^+}\frac{f(a)}{a}=\frac{\pi}{3}.$$

选(C).

4. (B). 由$\dfrac{1}{2}\leqslant x+y\leqslant 1$,得$[\ln(x+y)]^3\leqslant 0$,于是$I_1=\iint\limits_D[\ln(x+y)]^3\mathrm{d}x\,\mathrm{d}y\leqslant 0$;

当$\dfrac{1}{2}\leqslant x+y\leqslant 1$时,由$(x+y)^3>\sin^3(x+y)>0$,得$I_2>I_3>0$.

故$I_2>I_3>I_1$. 选(B).

5. (D). $\iint\limits_{x^2+y^2\leqslant t^2} f(t-\sqrt{x^2+y^2})\mathrm{d}x\,\mathrm{d}y=2\pi\int_0^t rf(t-r)\mathrm{d}r\xlongequal{t-r=u}2\pi\left[t\int_0^t f(u)\mathrm{d}u-\int_0^t uf(u)\mathrm{d}u\right]$,

$\int_0^{e^{t^2}-1}\dfrac{\sin u}{\sqrt{u}}\mathrm{d}u\sim\int_0^{t^2}\sqrt{u}\,\mathrm{d}u=\dfrac{2}{3}t^3(t\to 0^+)$,则

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$$\lim_{t\to0^+}\frac{\iint\limits_{x^2+y^2\leqslant t^2} f(t-\sqrt{x^2+y^2})\,dx\,dy}{\int_0^{e^{t^2}-1}\frac{\sin u}{\sqrt{u}}du}=3\pi\lim_{t\to0^+}\frac{t\int_0^t f(u)\,du-\int_0^t uf(u)\,du}{t^3}$$

$$=\pi\lim_{t\to0^+}\frac{\int_0^t f(u)\,du}{t^2}=\frac{\pi}{2}\lim_{t\to0^+}\frac{f(t)}{t}=\frac{\pi}{2}f'(0)=2\pi.$$

选(D).

6.(D). 因为积分区域 D 关于 $y=x$ 对称,所以

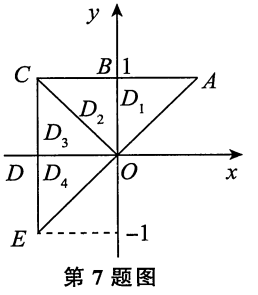
$$I=\iint_D\frac{af(x)+bf(y)}{f(x)+f(y)}dx\,dy=\iint_D\frac{af(y)+bf(x)}{f(y)+f(x)}dx\,dy,$$

$$2I=\iint_D\frac{af(x)+bf(y)}{f(x)+f(y)}dx\,dy+\iint_D\frac{af(y)+bf(x)}{f(y)+f(x)}dx\,dy$$

$$=(a+b)\iint_D dx\,dy=\frac{(a+b)}{4}\cdot(4\pi-\pi)=\frac{3(a+b)}{4}\pi,$$

故原式 $=\dfrac{3(a+b)}{8}\pi$. 选(D).

7.(A). 存在 5 个点分别为 $A(1,1),B(0,1),C(-1,1),D(-1,0),E(-1,-1)$,将 $\triangle OAB,\triangle OBC,\triangle OCD,\triangle ODE$ 所在的区域分别记为 D_1,D_2,D_3,D_4,如图所示.



第 7 题图

$$\iint_D(xy+\cos x\sin y)\,d\sigma=\left(\iint_{D_1}+\iint_{D_2}+\iint_{D_3}+\iint_{D_4}\right)(xy+\cos x\sin y)\,d\sigma,$$

根据对称性可知,

$$\left(\iint_{D_1}+\iint_{D_2}\right)(xy+\cos x\sin y)\,d\sigma=\left(\iint_{D_1}+\iint_{D_2}\right)\cos x\sin y\,d\sigma$$

$$=2\iint_{D_1}\cos x\sin y\,d\sigma=2\iint_{D_1}\cos x\sin y\,dx\,dy;$$

$$\left(\iint_{D_3}+\iint_{D_4}\right)(xy+\cos x\sin y)\,d\sigma=0.$$ 选(A).

8.(D). **方法一** 令 $F(x)=\int_a^x f(t)\,dt$,则 $F(a)=0,F(b)=A$,则

$$\int_a^b dy\int_y^b f(x)f(y)\,dx=\int_a^b f(y)\,dy\int_y^b f(x)\,dx=\int_a^b f(y)[F(b)-F(y)]dy$$

$$=F(b)\int_a^b f(y)\,dy-\int_a^b f(y)F(y)\,dy=A^2-\int_a^b F(y)\,d[F(y)]$$

$$=A^2-\frac{1}{2}F^2(y)\Big|_a^b=\frac{A^2}{2}.$$

选(D).

方法二 令 $I=\int_a^b dy\int_y^b f(x)f(y)\,dx$,由 $I=\int_a^b dx\int_a^x f(x)f(y)\,dy=\int_a^b dy\int_a^y f(y)f(x)\,dx$,得

$$2I=\int_a^b dy\int_a^b f(x)f(y)\,dx=\int_a^b f(x)\,dx\int_a^b f(y)\,dy=A^2,$$

则 $I=\dfrac{A^2}{2}$. 选(D).

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9. (A). 将区域向 y 轴投影. 令

$D_1=\{(x,y)\mid 0\leqslant x\leqslant y,0\leqslant y\leqslant 1\}$, $D_2=\{(x,y)\mid 0\leqslant x\leqslant\sqrt{2-y^2},1\leqslant y\leqslant\sqrt{2}\}$,

则 $\int_0^1 dx\int_x^{\sqrt{2-x^2}} f(x,y)dy=\int_0^1 dy\int_0^y f(x,y)dx+\int_1^{\sqrt{2}} dy\int_0^{\sqrt{2-y^2}} f(x,y)dx$. 选(A).

10. (B). 令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta,\end{cases}$ 其中 $0\leqslant\theta\leqslant\pi,0\leqslant r\leqslant 2a\sin\theta$.

则 $\iint\limits_D f(x,y)dx\,dy=\int_0^{\pi}d\theta\int_0^{2a\sin\theta} f(r\cos\theta,r\sin\theta)r\,dr$. 选(B).

11. (D). 令 $x=r\cos\theta,y=r\sin\theta$. 由 $0\leqslant\theta\leqslant\dfrac{\pi}{2}$,得 $y\geqslant 0$;又由 $0\leqslant r\leqslant 2\cos\theta$,得 $0\leqslant r^2\leqslant 2r\cos\theta$,即 $x^2+y^2\leqslant 2x$. 故累次积分所对应的二重积分的积分区域为 $D:x^2+y^2\leqslant 2x\,(y\geqslant 0)$,

则 $D=\{(x,y)\mid 0\leqslant x\leqslant 2,0\leqslant y\leqslant\sqrt{2x-x^2}\}$,故

$$\int_0^{\frac{\pi}{2}}d\theta\int_0^{2\cos\theta} f(r\cos\theta,r\sin\theta)r\,dr=\int_0^2 dx\int_0^{\sqrt{2x-x^2}} f(x,y)dy.$$

选(D).

12. (A). 改变积分次序,得

$$\int_0^2 dy\int_y^2 y\ln(1+x^3)dx=\int_0^2\ln(1+x^3)dx\int_0^x y\,dy=\frac{1}{2}\int_0^2 x^2\ln(1+x^3)dx$$

$$=\frac{1}{6}\int_0^2\ln(1+x^3)d(1+x^3)\xlongequal{1+x^3=t}\frac{1}{6}\int_1^9\ln t\,dt=\frac{1}{6}t\ln t\Big|_1^9-\frac{4}{3}=3\ln 3-\frac{4}{3}.$$

选(A).

二、填空题

1. 解 $\lim\limits_{n\to\infty}\dfrac{1}{n^2}\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{2n}\dfrac{ij}{n^2}\sin^2\dfrac{\pi j}{n}=\lim\limits_{n\to\infty}\dfrac{1}{n^2}\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{2n}\dfrac{i}{n}\cdot\dfrac{j}{n}\sin^2\dfrac{\pi j}{n}=\int_0^1 dx\int_0^2 xy\sin^2\pi y\,dy$

$$=\int_0^1 x\,dx\int_0^2 y\sin^2\pi y\,dy=\frac{1}{2\pi^2}\int_0^2(\pi y)\sin^2\pi y\,d(\pi y)=\frac{1}{2\pi^2}\int_0^{2\pi}x\sin^2 x\,dx$$

$$\xlongequal{x-\pi=t}\frac{1}{2\pi^2}\int_{-\pi}^{\pi}(\pi+t)\sin^2 t\,dt=\frac{1}{\pi}\int_0^{\pi}\sin^2 t\,dt=\frac{2}{\pi}\int_0^{\frac{\pi}{2}}\sin^2 t\,dt=\frac{2}{\pi}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=\frac{1}{2}.$$

2. 解 令 $D_1=\{(x,y)\mid 0\leqslant x\leqslant 1,0\leqslant y\leqslant x^2\}$,

$D_2=\{(x,y)\mid 1\leqslant x\leqslant\sqrt{2},0\leqslant y\leqslant\sqrt{2-x^2}\}$,

则 $\int_0^1 dy\int_{\sqrt{y}}^{\sqrt{2-y^2}} f(x,y)dx=\int_0^1 dx\int_0^{x^2} f(x,y)dy+\int_1^{\sqrt{2}}dx\int_0^{\sqrt{2-x^2}} f(x,y)dy$.

3. 解 改变积分次序,得

$$\int_0^1 dy\int_y^1\frac{y}{\sqrt{1+x^6}}dx=\int_0^1 dx\int_0^x\frac{y}{\sqrt{1+x^6}}dy=\frac{1}{2}\int_0^1\frac{x^2}{\sqrt{1+x^6}}dx$$

$$=\frac{1}{6}\int_0^1\frac{d(x^3)}{\sqrt{1+(x^3)^2}}=\frac{1}{6}\int_0^1\frac{dx}{\sqrt{1+x^2}}=\frac{1}{6}\ln(x+\sqrt{1+x^2})\Big|_0^1$$

$$=\frac{1}{6}\ln(1+\sqrt{2}).$$

4. 解 改变积分次序,得

$$\int_0^1 dy\int_{\sqrt{y}}^{y} e^{-\frac{y}{x}}dx=\int_0^1 dx\int_{x^2}^{x} e^{-\frac{y}{x}}dy=-\int_0^1 x\,e^{-\frac{y}{x}}\Big|_{x^2}^{x}dx$$

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$$=-\int_0^1 x(\mathrm{e}^{-1}-\mathrm{e}^{-x})\mathrm{d}x=-\frac{1}{2\mathrm{e}}+\int_0^1(-x)\mathrm{e}^{-x}\mathrm{d}(-x)$$

$$=-\frac{1}{2\mathrm{e}}-(x+1)\mathrm{e}^{-x}\Big|_0^1=-\frac{1}{2\mathrm{e}}-\left(\frac{2}{\mathrm{e}}-1\right)=1-\frac{5}{2\mathrm{e}}.$$

5. 解 改变积分次序,得

$$\int_0^{+\infty}x\mathrm{d}x\int_x^{+\infty}\mathrm{e}^{-y^2}\mathrm{d}y=\int_0^{+\infty}\mathrm{e}^{-y^2}\mathrm{d}y\int_0^y x\mathrm{d}x=\frac{1}{2}\int_0^{+\infty}y^2\mathrm{e}^{-y^2}\mathrm{d}y\xlongequal{y^2=t}\frac{1}{2}\int_0^{+\infty}t\mathrm{e}^{-t}\cdot\frac{1}{2\sqrt{t}}\mathrm{d}t$$

$$=\frac{1}{4}\int_0^{+\infty}t^{\frac{1}{2}}\mathrm{e}^{-t}\mathrm{d}t=\frac{1}{4}\Gamma\left(\frac{1}{2}+1\right)=\frac{\sqrt{\pi}}{8}.$$

6. 解 令$\iint\limits_D f(x,y)\mathrm{d}\sigma=k$,则 $f(x,y)=xy+k$,等式两边在 D 上积分,得$\iint\limits_D f(x,y)\mathrm{d}\sigma=\iint\limits_D(xy+k)\mathrm{d}\sigma$,即 $k=\int_0^1\mathrm{d}x\int_0^{x^2}(xy+k)\mathrm{d}y$,解得 $k=\frac{1}{8}$,所以 $f(x,y)=xy+\frac{1}{8}$.

三、解答题

1. 解 (1) 因为 $D=\{(x,y)\mid a-\sqrt{a^2-y^2}\leqslant x\leqslant y,0\leqslant y\leqslant a\}$,所以

$$\int_0^a\mathrm{d}x\int_x^{\sqrt{2ax-x^2}}f(x,y)\mathrm{d}y=\int_0^a\mathrm{d}y\int_{a-\sqrt{a^2-y^2}}^y f(x,y)\mathrm{d}x.$$

(2) $D=\left\{(x,y)\,\middle|\,0\leqslant x\leqslant\frac{1}{2},x^2\leqslant y\leqslant x\right\}$,则

$$\int_0^{\frac{1}{4}}\mathrm{d}y\int_y^{\sqrt{y}}f(x,y)\mathrm{d}x+\int_{\frac{1}{4}}^{\frac{1}{2}}\mathrm{d}y\int_y^{\frac{1}{2}}f(x,y)\mathrm{d}x=\int_0^{\frac{1}{2}}\mathrm{d}x\int_{x^2}^x f(x,y)\mathrm{d}y.$$

2. 解 令$\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(\frac{\pi}{4}\leqslant\theta\leqslant\frac{3\pi}{4},0\leqslant r\leqslant\csc\theta\right)$,则

$$\int_0^1\mathrm{d}y\int_{-y}^y f(x,y)\mathrm{d}x=\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\mathrm{d}\theta\int_0^{\csc\theta}rf(r\cos\theta,r\sin\theta)\mathrm{d}r.$$

3. 解 (1) 令 $D_1=\{(x,y)\mid 1\leqslant x\leqslant 2,\sqrt{x}\leqslant y\leqslant x\}$,$D_2=\{(x,y)\mid 2\leqslant x\leqslant 4,\sqrt{x}\leqslant y\leqslant 2\}$,$D_1+D_2=D=\{(x,y)\mid 1\leqslant y\leqslant 2,y\leqslant x\leqslant y^2\}$,则

$$\int_1^2\mathrm{d}x\int_{\sqrt{x}}^x\sin\frac{\pi x}{2y}\mathrm{d}y+\int_2^4\mathrm{d}x\int_{\sqrt{x}}^2\sin\frac{\pi x}{2y}\mathrm{d}y=\int_1^2\mathrm{d}y\int_y^{y^2}\sin\frac{\pi x}{2y}\mathrm{d}x$$

$$=-\frac{2}{\pi}\int_1^2 y\cos\frac{\pi}{2}y\mathrm{d}y=-\frac{8}{\pi^3}\int_1^2\frac{\pi}{2}y\cos\frac{\pi}{2}y\mathrm{d}\left(\frac{\pi}{2}y\right)$$

$$=-\frac{8}{\pi^3}\int_{\frac{\pi}{2}}^{\pi}t\cos t\mathrm{d}t=-\frac{8}{\pi^3}\left(t\sin t\Big|_{\frac{\pi}{2}}^{\pi}-\int_{\frac{\pi}{2}}^{\pi}\sin t\mathrm{d}t\right)=\frac{4}{\pi^3}(\pi+2).$$

(2) 改变积分次序,得

$$\int_0^2\mathrm{d}y\int_y^2 x^2\mathrm{e}^{-x^2}\mathrm{d}x=\int_0^2\mathrm{d}x\int_0^x x^2\mathrm{e}^{-x^2}\mathrm{d}y=\int_0^2 x^3\mathrm{e}^{-x^2}\mathrm{d}x$$

$$=\frac{1}{2}\int_0^2 x^2\mathrm{e}^{-x^2}\mathrm{d}(x^2)=\frac{1}{2}\int_0^4 x\mathrm{e}^{-x}\mathrm{d}x$$

$$=-\frac{1}{2}(x+1)\mathrm{e}^{-x}\Big|_0^4=\frac{1}{2}-\frac{5}{2\mathrm{e}^4}.$$

4. 解 令 $x=r\cos\theta,y=r\sin\theta$,则 $F(t)=\int_0^{2\pi}\mathrm{d}\theta\int_0^t rf(r^2)\mathrm{d}r=2\pi\int_0^t rf(r^2)\mathrm{d}r$.

因为 $f(x)$ 连续,所以 $F'(t)=2\pi tf(t^2)$ 且 $F'(0)=0$,于是

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$$F''(0)=\lim_{t\to0^+}\frac{F'(t)-F'(0)}{t}=\lim_{t\to0^+}2\pi f(t^2)=2\pi f(0)=2\pi.$$

5. 解 (1) $\iint\limits_D xy(x+y)\mathrm{d}\sigma=\iint\limits_D x^2y\mathrm{d}\sigma+\iint\limits_D xy^2\mathrm{d}\sigma$

$$=\iint\limits_D x^2y\mathrm{d}\sigma=2\int_0^1 y\mathrm{d}y\int_0^{\sqrt{1+y^2}}x^2\mathrm{d}x$$

$$=\frac{2}{3}\int_0^1 y(1+y^2)^{\frac{3}{2}}\mathrm{d}y=\frac{1}{3}\int_0^1(1+y^2)^{\frac{3}{2}}\mathrm{d}(1+y^2)$$

$$=\frac{2}{15}(1+y^2)^{\frac{5}{2}}\Big|_0^1=\frac{2}{15}(4\sqrt{2}-1).$$

(2) 令 $D_0=\{(x,y)\mid 0\leqslant x\leqslant 1,0\leqslant y\leqslant 1\}$，则 $\iint\limits_D\sqrt{|x-|y||}\mathrm{d}\sigma=2\iint\limits_{D_0}\sqrt{|x-y|}\mathrm{d}\sigma$，

而 $\iint\limits_{D_0}\sqrt{|x-y|}\mathrm{d}\sigma=2\int_0^1\mathrm{d}x\int_0^x\sqrt{x-y}\mathrm{d}y=\frac{4}{3}\int_0^1x^{\frac{3}{2}}\mathrm{d}x=\frac{8}{15}$，故 $\iint\limits_D\sqrt{|x-|y||}\mathrm{d}\sigma=\frac{16}{15}$.

(3) 由 $\sqrt{\frac{x}{a}}+\sqrt{\frac{y}{b}}=1$，得 $y=b\left(1-\sqrt{\frac{x}{a}}\right)^2$，因此

$$I=\iint\limits_D y\mathrm{d}x\mathrm{d}y=\int_0^a\mathrm{d}x\int_0^{b\left(1-\sqrt{\frac{x}{a}}\right)^2}y\mathrm{d}y=\frac{b^2}{2}\int_0^a\left(1-\sqrt{\frac{x}{a}}\right)^4\mathrm{d}x.$$

令 $t=1-\sqrt{\frac{x}{a}}$，则 $x=a(1-t)^2$，$\mathrm{d}x=-2a(1-t)\mathrm{d}t$，于是

$$I=\frac{b^2}{2}\int_0^a\left(1-\sqrt{\frac{x}{a}}\right)^4\mathrm{d}x=ab^2\int_0^1(t^4-t^5)\mathrm{d}t=\frac{ab^2}{30}.$$

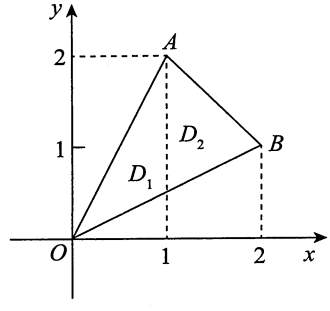
(4) 将区域向 x 轴投影，如图所示，令

$$D_1=\left\{(x,y)\,\middle|\,0\leqslant x\leqslant 1,\frac{x}{2}\leqslant y\leqslant 2x\right\},$$

$$D_2=\left\{(x,y)\,\middle|\,1\leqslant x\leqslant 2,\frac{x}{2}\leqslant y\leqslant 3-x\right\},$$

则 $\iint\limits_D x\mathrm{d}x\mathrm{d}y=\int_0^1x\mathrm{d}x\int_{\frac{x}{2}}^{2x}\mathrm{d}y+\int_1^2x\mathrm{d}x\int_{\frac{x}{2}}^{3-x}\mathrm{d}y$

$$=\int_0^1\frac{3x^2}{2}\mathrm{d}x+\int_1^2\left(3x-\frac{3x^2}{2}\right)\mathrm{d}x=\frac{3}{2}.$$



第 5-(4) 题图

(5) 如图所示，直线 $x+y=\frac{\pi}{2}$ 将区域 D 分为 D_1,D_2，其中

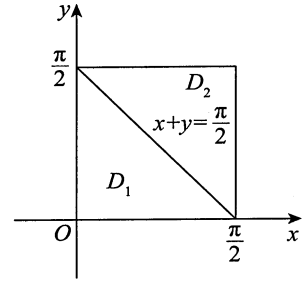
$$D_1=\left\{(x,y)\,\middle|\,0\leqslant x\leqslant\frac{\pi}{2},0\leqslant y\leqslant\frac{\pi}{2}-x\right\},$$

$$D_2=\left\{(x,y)\,\middle|\,0\leqslant x\leqslant\frac{\pi}{2},\frac{\pi}{2}-x\leqslant y\leqslant\frac{\pi}{2}\right\},$$

则 $I=\iint\limits_D|\cos(x+y)|\mathrm{d}x\mathrm{d}y$

$$=\iint\limits_{D_1}\cos(x+y)\mathrm{d}x\mathrm{d}y-\iint\limits_{D_2}\cos(x+y)\mathrm{d}x\mathrm{d}y,$$

其中



第 5-(5) 题图

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$$\iint_{D_1}\cos(x+y)\,dx\,dy=\int_0^{\frac{\pi}{2}}dx\int_0^{\frac{\pi}{2}-x}\cos(x+y)\,dy=\int_0^{\frac{\pi}{2}}(1-\sin x)\,dx=\frac{\pi}{2}-1,$$

$$\iint_{D_2}\cos(x+y)\,dx\,dy=\int_0^{\frac{\pi}{2}}dx\int_{\frac{\pi}{2}-x}^{\frac{\pi}{2}}\cos(x+y)\,dy=\int_0^{\frac{\pi}{2}}(\cos x-1)\,dx=1-\frac{\pi}{2},$$

故 $I=\iint_D|\cos(x+y)|\,dx\,dy=\pi-2.$

6. 解 (1) 由极坐标法得

$$\iint_D\frac{1-x^2-y^2}{1+x^2+y^2}dx\,dy=\int_0^{\frac{\pi}{2}}d\theta\int_0^1\frac{r(1-r^2)}{1+r^2}dr=\frac{\pi}{4}\int_0^1\frac{1-r^2}{1+r^2}d(r^2)$$

$$\xlongequal{r^2=t}\frac{\pi}{4}\int_0^1\frac{1-t}{1+t}dt=\frac{\pi}{4}\int_0^1\left(-1+\frac{2}{1+t}\right)dt=\frac{\pi}{4}\ln\frac{4}{e}.$$

(2) **方法一** 由积分区域关于 $y=x$ 对称,可得

$$I=\iint_D\frac{x\sin(x^2+y^2)}{\sqrt{x^2+y^2}}dx\,dy=\frac{1}{2}\iint_D\frac{(x+y)\sin(x^2+y^2)}{\sqrt{x^2+y^2}}dx\,dy$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}(\sin\theta+\cos\theta)\,d\theta\int_0^1 r\sin r^2\,dr=-\frac{1}{2}\cos r^2\Big|_0^1=\frac{1-\cos 1}{2}.$$

方法二 $I=\iint_D\frac{x\sin(x^2+y^2)}{\sqrt{x^2+y^2}}dx\,dy=\int_0^{\frac{\pi}{2}}\cos\theta\,d\theta\int_0^1 r\sin r^2\,dr=\int_0^1 r\sin r^2\,dr=\frac{1-\cos 1}{2}.$

(3) 由积分区域关于 $y=x$ 对称,得 $I=\iint_D\sin x^2\cos y^2\,dx\,dy=\iint_D\sin y^2\cos x^2\,dx\,dy$,则

$$2I=\iint_D\sin x^2\cos y^2\,dx\,dy+\iint_D\sin y^2\cos x^2\,dx\,dy=\iint_D\sin(x^2+y^2)\,dx\,dy$$

$$=\int_0^{\frac{\pi}{2}}d\theta\int_0^a r\sin r^2\,dr=-\frac{\pi}{4}\cos r^2\Big|_0^a=\frac{\pi}{4}(1-\cos a^2).$$

故 $I=\frac{\pi}{8}(1-\cos a^2).$

7. 解 (1) 令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(-\frac{\pi}{4}\leqslant\theta\leqslant 0,0\leqslant r\leqslant -2a\sin\theta\right)$,则

$$I=\iint_D\frac{\sqrt{x^2+y^2}}{\sqrt{4a^2-x^2-y^2}}dx\,dy=\int_{-\frac{\pi}{4}}^0 d\theta\int_0^{-2a\sin\theta}\frac{r^2}{\sqrt{4a^2-r^2}}dr$$

$$\xlongequal{r=2a\sin t}\int_{-\frac{\pi}{4}}^0 d\theta\int_0^{-\theta}\frac{4a^2\sin^2 t}{2a\cos t}\cdot 2a\cos t\,dt=2a^2\int_{-\frac{\pi}{4}}^0 d\theta\int_0^{-\theta}(1-\cos 2t)\,dt$$

$$=2a^2\int_{-\frac{\pi}{4}}^0\left(-\theta+\frac{1}{2}\sin 2\theta\right)d\theta=a^2\left(\frac{\pi^2}{16}-\frac{1}{2}\right).$$

(2) 令 $\begin{cases}x-\frac{1}{2}=r\cos\theta,\\ y-\frac{1}{2}=r\sin\theta\end{cases}\left(0\leqslant\theta\leqslant 2\pi,0\leqslant r\leqslant\frac{\sqrt{6}}{2}\right)$,则

$$\iint_D(x+y)\,dx\,dy=\int_0^{2\pi}d\theta\int_0^{\frac{\sqrt{6}}{2}}(1+r\cos\theta+r\sin\theta)\,r\,dr$$

$$=\int_0^{2\pi}\left(\frac{3}{4}+\frac{\sqrt{6}}{4}\cos\theta+\frac{\sqrt{6}}{4}\sin\theta\right)d\theta=\frac{3\pi}{2}.$$

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(3) 令$\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(0\leqslant\theta\leqslant\dfrac{\pi}{2},0\leqslant r\leqslant 2a\cos\theta\right)$,则

$$\int_0^{2a}\mathrm{d}x\int_0^{\sqrt{2ax-x^2}}(x+y)^2\mathrm{d}y=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{2a\cos\theta}r^3(\sin\theta+\cos\theta)^2\mathrm{d}r$$

$$=4a^4\int_0^{\frac{\pi}{2}}(1+2\sin\theta\cos\theta)\cos^4\theta\,\mathrm{d}\theta$$

$$=4a^4\int_0^{\frac{\pi}{2}}\cos^4\theta\,\mathrm{d}\theta+8a^4\int_0^{\frac{\pi}{2}}\sin\theta\cos^5\theta\,\mathrm{d}\theta$$

$$=4a^4\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}-\frac{4}{3}a^4\cos^6\theta\Big|_0^{\frac{\pi}{2}}=\left(\frac{3\pi}{4}+\frac{4}{3}\right)a^4.$$

(4) 令$\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(0\leqslant\theta\leqslant\dfrac{\pi}{4},0\leqslant r\leqslant 4\cos\theta\right)$,则

$$\iint_D(x^2+y^2)\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_0^{4\cos\theta}r^3\mathrm{d}r=64\int_0^{\frac{\pi}{4}}\cos^4\theta\,\mathrm{d}\theta=16\int_0^{\frac{\pi}{4}}(1+\cos 2\theta)^2\mathrm{d}\theta$$

$$=8\int_0^{\frac{\pi}{2}}(1+\cos t)^2\mathrm{d}t=8\int_0^{\frac{\pi}{2}}(1+2\cos t+\cos^2 t)\mathrm{d}t$$

$$=8\left(\frac{\pi}{2}+2+\frac{\pi}{4}\right)=6\pi+16.$$

(5)$\iint\limits_D(x+y)^2\mathrm{d}x\,\mathrm{d}y=\iint\limits_D(x^2+2xy+y^2)\mathrm{d}x\,\mathrm{d}y=\iint\limits_D(x^2+y^2)\mathrm{d}x\,\mathrm{d}y.$

令$\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}(0\leqslant\theta\leqslant\pi,a\sin\theta\leqslant r\leqslant 2a\sin\theta)$,则

$$\text{原式}=\int_0^{\pi}\mathrm{d}\theta\int_{a\sin\theta}^{2a\sin\theta}r^3\mathrm{d}r=\frac{15}{4}a^4\int_0^{\pi}\sin^4\theta\,\mathrm{d}\theta=\frac{15}{2}a^4\int_0^{\frac{\pi}{2}}\sin^4\theta\,\mathrm{d}\theta$$

$$=\frac{15}{2}a^4\cdot I_4=\frac{15}{2}a^4\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=\frac{45\pi a^4}{32}.$$

8. 解 **方法一** $I=\int_0^2\mathrm{d}y\int_{-2}^{-\sqrt{2y-y^2}}y^2\mathrm{d}x=\int_0^2y^2(2-\sqrt{2y-y^2})\mathrm{d}y$

$$=2\int_0^2y^2\mathrm{d}y-\int_0^2y^2\sqrt{1-(y-1)^2}\,\mathrm{d}y$$

$$\xlongequal{y=t+1}\frac{16}{3}-\int_{-1}^1(t+1)^2\sqrt{1-t^2}\,\mathrm{d}t$$

$$=\frac{16}{3}-2\int_0^1t^2\sqrt{1-t^2}\,\mathrm{d}t-\frac{\pi}{2}$$

$$=\frac{16}{3}-\frac{\pi}{2}-2\int_0^{\frac{\pi}{2}}\sin^2\theta\cos^2\theta\,\mathrm{d}\theta=\frac{16}{3}-\frac{5\pi}{8}.$$

方法二 由$x=-\sqrt{2y-y^2}$及y轴围成的区域为D_1,则$I=\left(\iint\limits_{D+D_1}-\iint\limits_{D_1}\right)y^2\mathrm{d}\sigma=I_1-I_2$,而

$$I_1=\iint_{D+D_1}y^2\mathrm{d}\sigma=\int_{-2}^0\mathrm{d}x\int_0^2y^2\mathrm{d}y=\frac{16}{3},$$

$$I_2=\int_{\frac{\pi}{2}}^{\pi}\mathrm{d}\theta\int_0^{2\sin\theta}r^3\sin^2\theta\,\mathrm{d}r=4\int_0^{\frac{\pi}{2}}\sin^6\theta\,\mathrm{d}\theta=\frac{5\pi}{8},$$

所以$I=\frac{16}{3}-\frac{5\pi}{8}$.

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9. 解 令 $D_0=\{(x,y)\mid x\geqslant 0,x^2+y^2\leqslant 2y\}$,则

$$\iint_D |x^2+y^2-2y|\,d\sigma=\iint_{D_0}(2y-x^2-y^2)d\sigma+\iint_{D-D_0}(x^2+y^2-2y)d\sigma$$

$$=2\iint_{D_0}(2y-x^2-y^2)d\sigma+\iint_D(x^2+y^2-2y)d\sigma,$$

而 $\iint_{D_0}(2y-x^2-y^2)d\sigma=\iint\limits_{\substack{x^2+(y-1)^2\leqslant 1\\ x\geqslant 0}}[1-x^2-(y-1)^2]dx\,d(y-1)$

$$=\iint\limits_{\substack{x^2+y^2\leqslant 1\\ x\geqslant 0}}(1-x^2-y^2)dx\,dy=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\int_0^1 r(1-r^2)dr=\frac{\pi}{4};$$

$$\iint_D(x^2+y^2-2y)d\sigma=\int_0^1 dx\int_0^1(x^2+y^2-2y)dy=\int_0^1\left(x^2-\frac{2}{3}\right)dx=-\frac{1}{3}.$$

故原式 $=\frac{\pi}{2}-\frac{1}{3}$.

10. 解 由对称性,得 $I=\iint_D\frac{y}{\sqrt{x^2+y^2}}dx\,dy=\frac{1}{2}\iint_D\frac{x+y}{\sqrt{x^2+y^2}}dx\,dy$.

令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(0\leqslant\theta\leqslant\frac{\pi}{2},\frac{1}{\cos\theta+\sin\theta}\leqslant r\leqslant\frac{2}{\cos\theta+\sin\theta}\right)$,则

$$I=\frac{1}{2}\int_0^{\frac{\pi}{2}}(\cos\theta+\sin\theta)d\theta\int_{\frac{1}{\cos\theta+\sin\theta}}^{\frac{2}{\cos\theta+\sin\theta}}r\,dr=\frac{3}{4}\int_0^{\frac{\pi}{2}}\frac{1}{\cos\theta+\sin\theta}d\theta$$

$$=\frac{3}{4\sqrt{2}}\int_0^{\frac{\pi}{2}}\frac{1}{\cos\left(\theta-\frac{\pi}{4}\right)}d\left(\theta-\frac{\pi}{4}\right)=\frac{3}{4\sqrt{2}}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec\theta\,d\theta=\frac{3}{2\sqrt{2}}\int_0^{\frac{\pi}{4}}\sec\theta\,d\theta$$

$$=\frac{3}{2\sqrt{2}}\ln|\sec\theta+\tan\theta|\Big|_0^{\frac{\pi}{4}}=\frac{3}{2\sqrt{2}}\ln(\sqrt{2}+1).$$

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